

Emon Roy Bappy

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Research Interest

- Electromechanical Systems
- Soft Robotics
- Bio-Inspired Robotics
- Automobile Engineering
- Additive Manufacturing
- Composite Materials
- Machine Design
- Computational Analysis

Education

Bangladesh University of Engineering and Technology

Dhaka, Bangladesh

B.Sc. in Mechanical Engineering

Sept 2018 – July 2024

- **CGPA:** 3.43 / 4.00
- **Undergraduate Thesis:** Design and Fabrication of a Tendon-Driven Soft Actuator for Surgical Operation
Supervisor: Prof. Dr. Md. Afsar Ali
- **Relevant Coursework:** Machine Design, Electromechanical System Design, Mechatronics, Composite Materials, Automobile Engineering, Bio-Engineering, Instrumentation and Measurement, Heat Transfer Equipment Design, Fluid Machinery

Research Experience

- **Biomimetic Systems:** Engineered a bio-inspired underwater robot with buoyancy control and maneuvering capabilities under the supervision of Dr. Rahman.
- **Industrial Robotics:** Three years of experience in designing and manufacturing medium- and small-platform Remotely Operated Vehicles (ROVs) with a 6-DoF robotic arm for industrial maintenance.
- **Machine Design:** Leading a research team to design and manufacture a double-enveloping worm gearbox, achieving 6 times the tooth contact area compared to conventional configurations.

Notable Projects

Man-Transportable Remotely Operated Vehicle (ROV)

Aug 2025 – Present

Professional Project

- Incorporated carbon-fiber body for lightweight design and track-belt drive with flipper arms, facilitating rough-terrain maneuverability, stair climbing and stability.
- Engineered a 7 DoF robotic arm with screw-based telescopic beam, capable of lifting 5kg payload at 1250mm extended position.

A Universal Soft Actuator for Surgical Applications

Apr 2023 – Jun 2024

Undergraduate Thesis

- Presented a flexible surgical actuator made of bio-compatible silicone rubber that reduces the number of incisions needed during minimal invasive surgery (MIS), speeds up recovery after surgery.
- Fabricated the actuator providing three degrees of freedom—bending along two axes and rotation at tip—enabling access to any location within its range.

Hilsha Bot, A Bio-Inspired Soft Robot

May 2022 – Sep 2022

Undergraduate Project - Electromechanical System Design

- Engineered an underwater electromechanical system, capable of moving like a fish with its soft tail actuator, and controlling its depth utilizing a buoyancy control system.
- Developed an appliance for underwater investigation, marine environment protection, and inspection of any underwater construction.

Titan, A Large Scale FDM Printer

Mar 2025 – Oct 2025

Professional Project, delivered to an architecture firm

- Constructed an FDM printer with 500mm×500mm×500mm build volume, CoreXY motion and a direct-drive dual-gear extruder.
- Manufactured the frame, build plate and motion system which facilitates 50–400 μm layer resolution and print speed up to 200 mm/s.

Anweshan and Renaissance, Mars Rovers for University Mars Rover Competition

Jun 2021 – Aug 2023

Undergraduate Team Project

- Built two Mars Rovers capable of traversing through rocky terrain, collecting and processing specimens from the environment, and performing maintenance tasks.
- Designed and implemented the PCB for the rover's power distribution system, actuators, and sensors for onboard scientific experiment.
- Led the electrical subteam in developing a 6 DoF robotic arm from scratch, capable of holding a 5 kg payload in extended position.

Formula Student Vehicle, A Combustion Vehicle for Formula Student Competition

Dec 2022 – Jul 2024

Undergraduate Team Project

- Led design and manufacturing process of a double wishbone suspension system with upright, bell crank, control arm and wheel hub.
- Performed analytical calculations for optimal spring rate, roll center and control arm geometry, along with stress analysis of suspension components including control arm, upright and bell crank.

Professional Experience

Cybernetics Hi-Tech Solutions (Pvt.) Ltd.
Research and Development Engineer

Dhaka, Bangladesh
01 July 2024 – Present

- Developed a man-portable ROV featuring a 7-DoF robotic arm actuated by integrated servo motor with harmonic gearbox and a custom double-enveloping worm gearbox.
- Analyzed the structural performance of fixed-wing UAVs and multirotor drones designed to carry high payloads of up to 50 kg.
- Acquired hands-on experience in manufacturing processes like composite manufacturing, wire EDM, CNC machining, fused deposition modeling, and molding.

Spectrum Engineering Consortium (Pvt.) Ltd.
Intern, Research and Development Team

Dhaka, Bangladesh
01 June 2023 – 30 June 2024

- Engineered an ROV with a 6 DoF arm, meticulously designed to operate and sustain functionality in harsh and challenging environments, ensuring robustness and reliability.
- Verified the structural integrity of the mechanical designs of an ROV through finite element analysis (FEA), identifying potential stress points and weaknesses.

Runner Automobiles PLC
Intern, 3 Wheeler Manufacturing Section

Mymensingh, Bangladesh
01 Nov 2023 – 20 Nov 2023

- Acquired comprehensive knowledge of three-wheeler manufacturing process, including welding, painting, and assembly sections.
- Developed a thorough understanding of automatic and semi-automatic line assembly and quality control, along with spot-welding automation using industrial robotic arms.
- Conducted a detailed analysis to identify process drawbacks and recommended improvements to enhance manufacturing efficiency.

Skills

Manufacturing Process	Composite Manufacturing(Resin Infusion, Hand Layup), FDM, Molding, CNC Milling, Wire EDM.
Computational Analysis	Ansys Mechanical, Structural Optimization, Ansys Composite PrePost.
Computer-Aided Design	SolidWorks, Ansys SpaceClaim, Ansys DesignModeler, AutoCAD, Blender.
Printer Circuit Board Design	Altium PCB Designer, Proteus, EasyEDA.
Programming Language	C, Python, Matlab.
Robotics	Arduino IDE, Robot Operating System (ROS).
Miscellaneous	Ubuntu, L ^A T _E X, Git.

Achievements

Finalist, University Rover Challenge, 2022.	The Mars Society
8th Position, European Rover Challenge Onsite, 2022.	European Rover Challenge(ERC)
Merit Award, BASIS National ICT Awards, 2022.	BASIS
Champion, Sustainable Energy Innovation Challenge, 2021.	Bikiron
Champion, National STEM Competition, 2021.	bdSTEM
4th place, 2nd Kibo Robot Programming Challenge Preliminary Round, 2021.	JAXA
Innovation Award, International Planetary Aerial Systems Challenge, 2021.	Mars Society South Asia
8th position, National Round of 6th Bangladesh Physics Olympiad, 2016.	BdPhO

Professional Certifications

Topology Optimization Using Ansys Mechanical	Ansys
Stress Analysis in Solid Mechanics	Ansys
SOLIDWORKS CAD Design Professional (CSWP)	Dassault Systèmes
Mechanical Design Solidworks Associate (CSWA)	Dassault Systèmes

References

Dr. Md. Afsar Ali Professor, Department of Mechanical Engineering Bangladesh University of Engineering and Technology (BUET) Email: mdaafsarali@me.buet.ac.bd Website: www.me.buet.ac.bd/faculty/dr-md-afsar-ali	Dr. Md Aman Uddin Assistant Professor, Department of Mechanical Engineering Bangladesh University of Engineering and Technology (BUET) Email: amanuddin@me.buet.ac.bd Website: https://me.buet.ac.bd/faculty/dr-md-aman-uddin-1
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